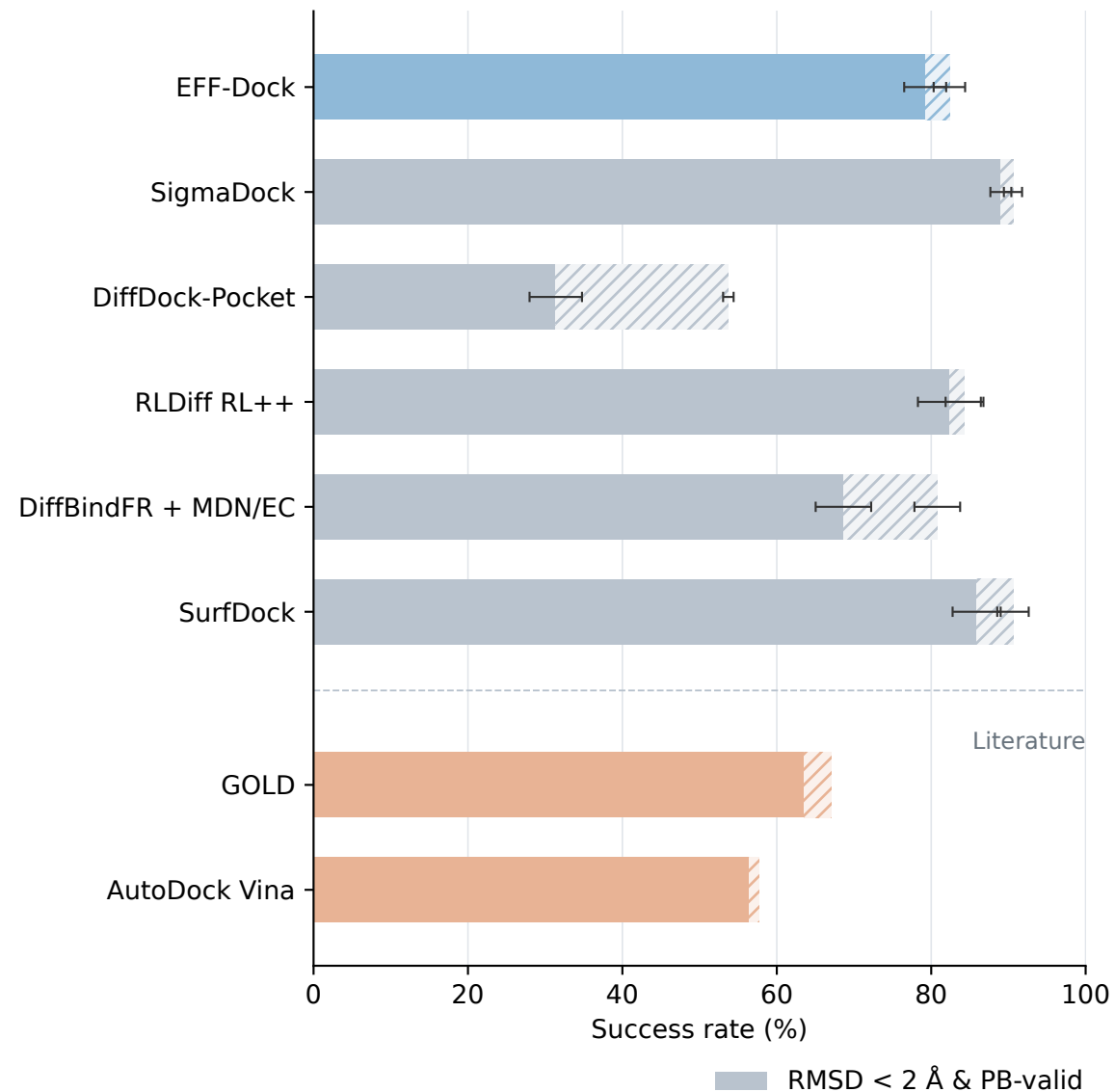
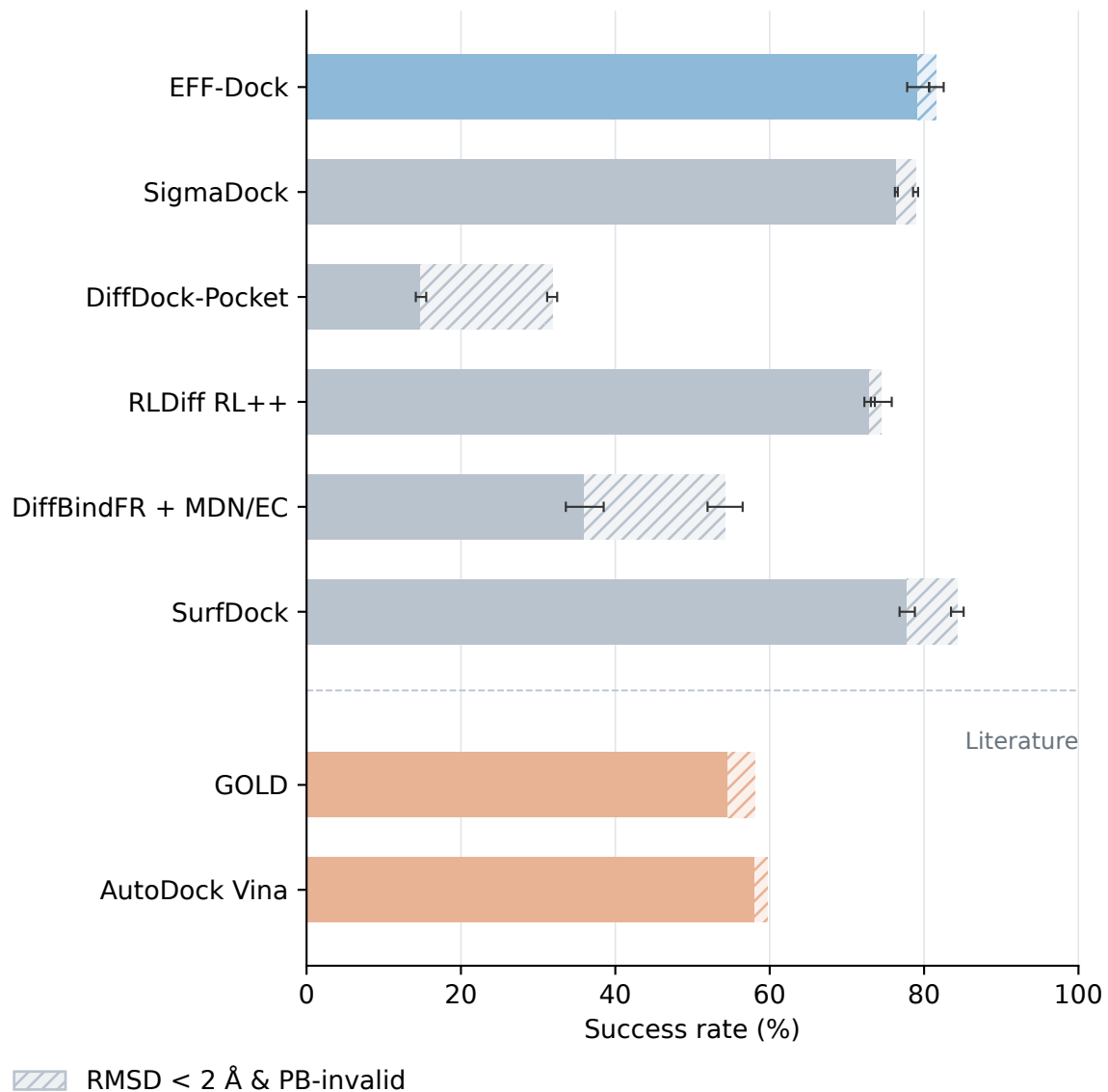
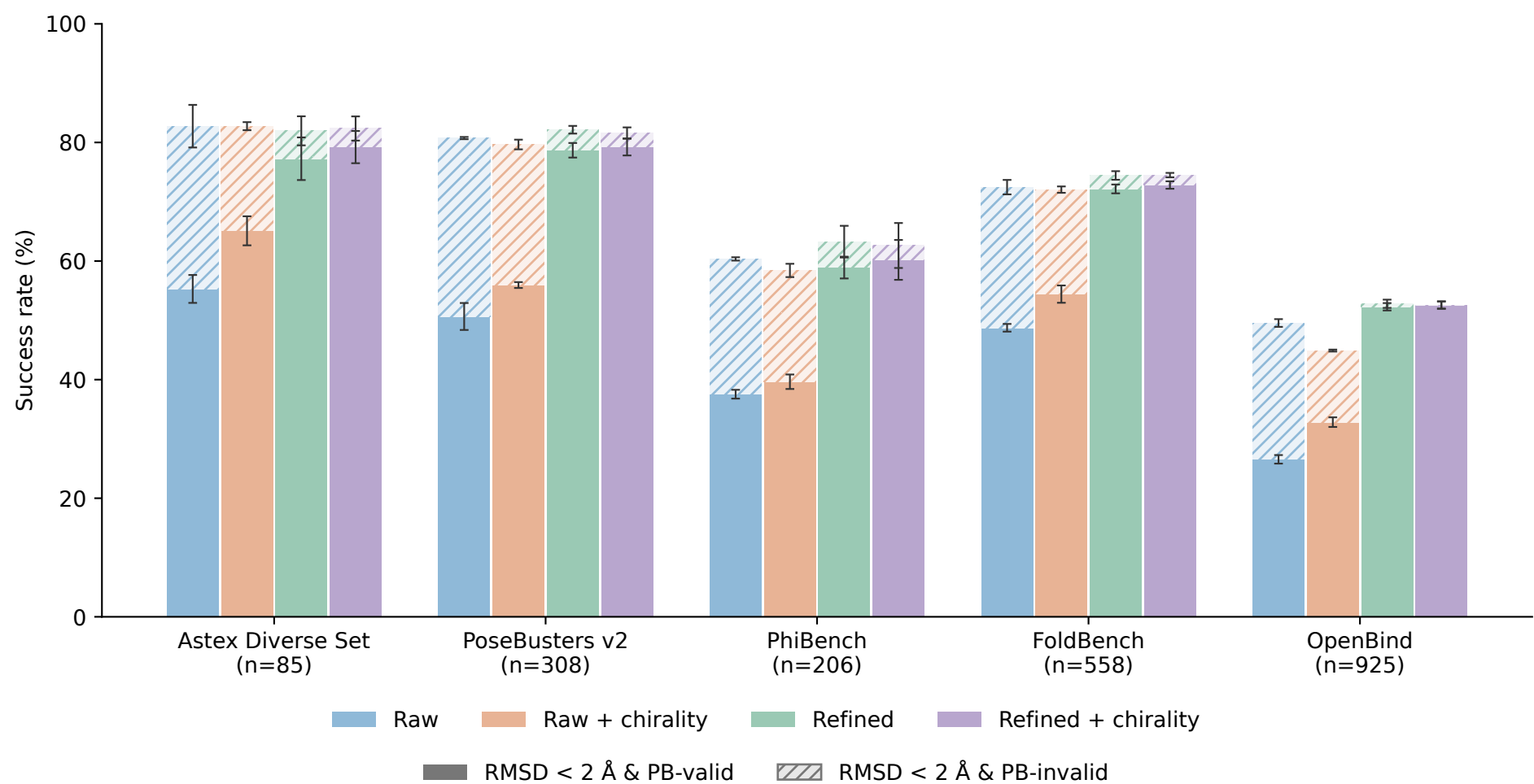
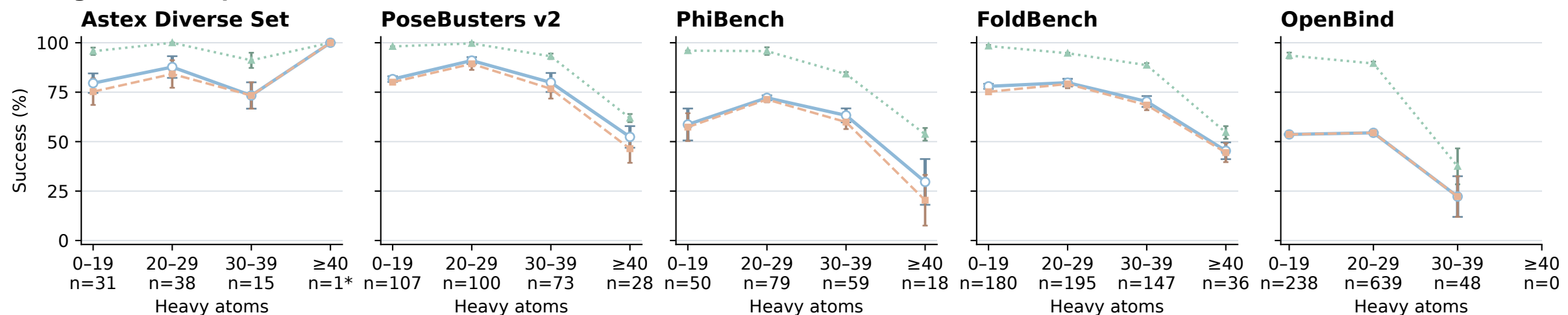


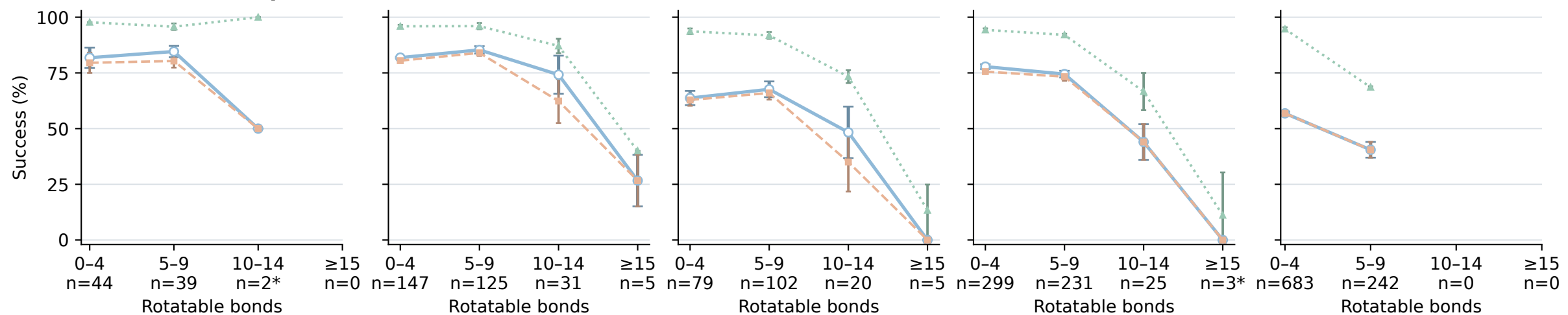
A Astex Diverse Set**B PoseBusters v2**



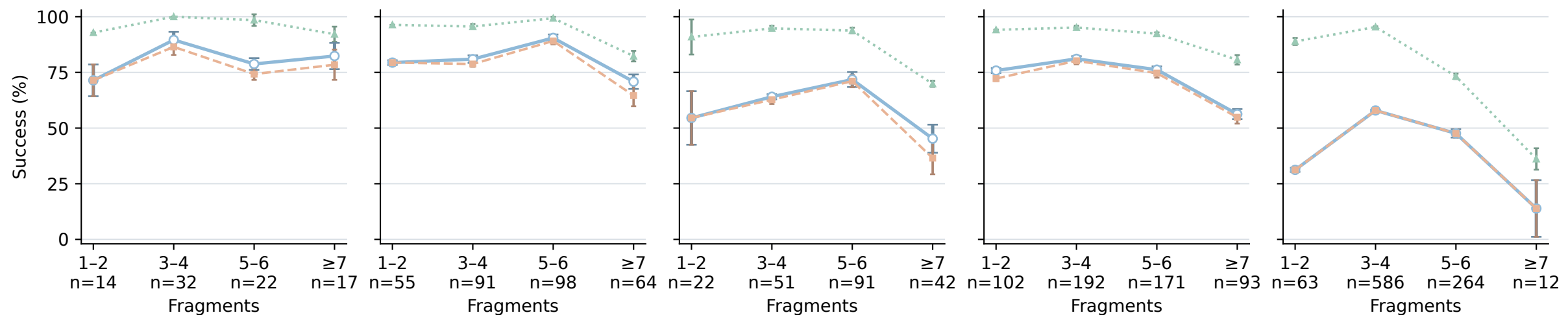
A Ligand size dependence



B Rotatable bond dependence

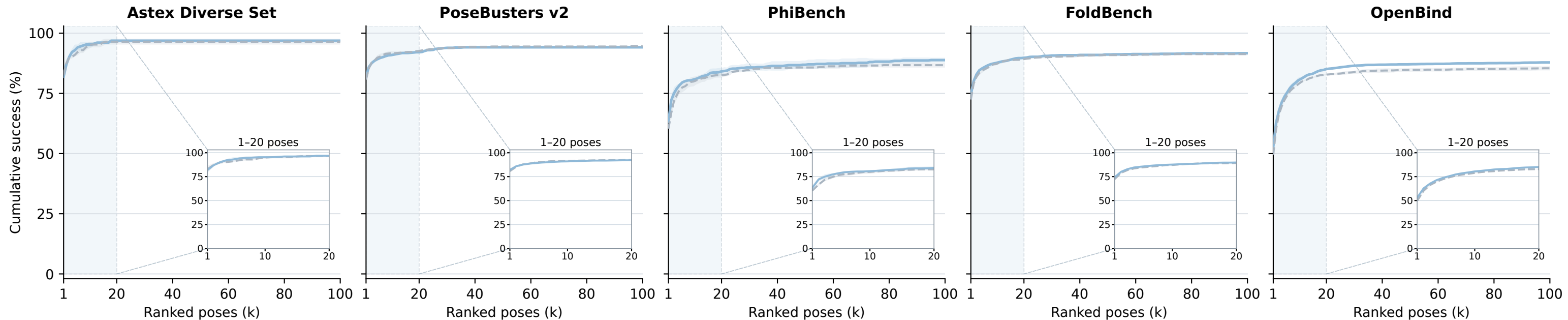


C Fragment count dependence

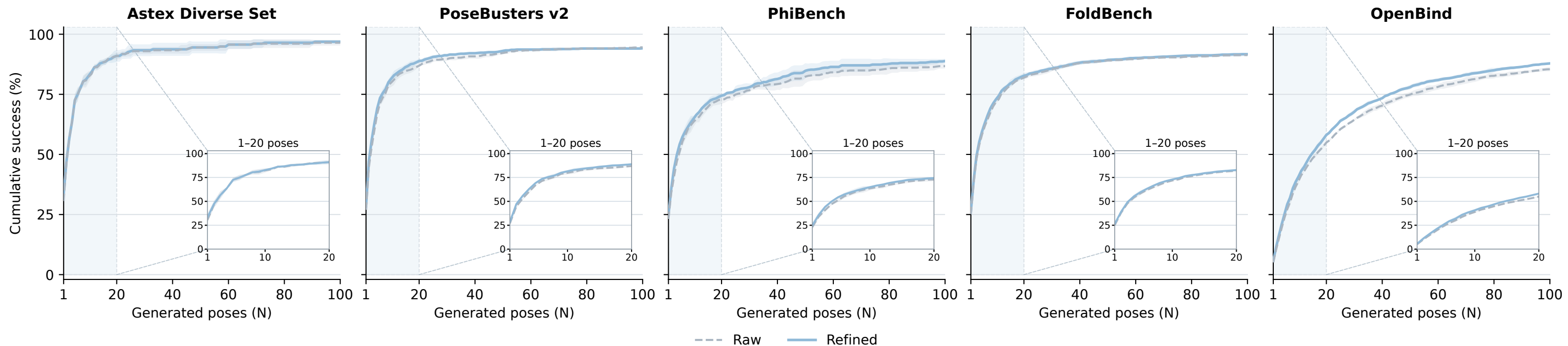


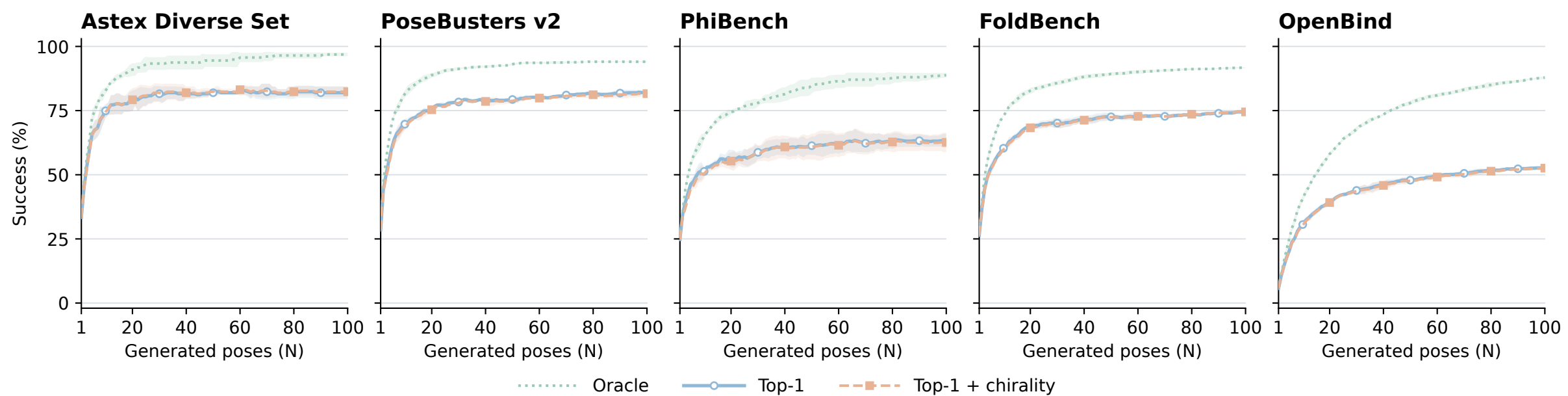
Oracle Top-1 Top-1 + PB-valid

A Confidence-ranked poses

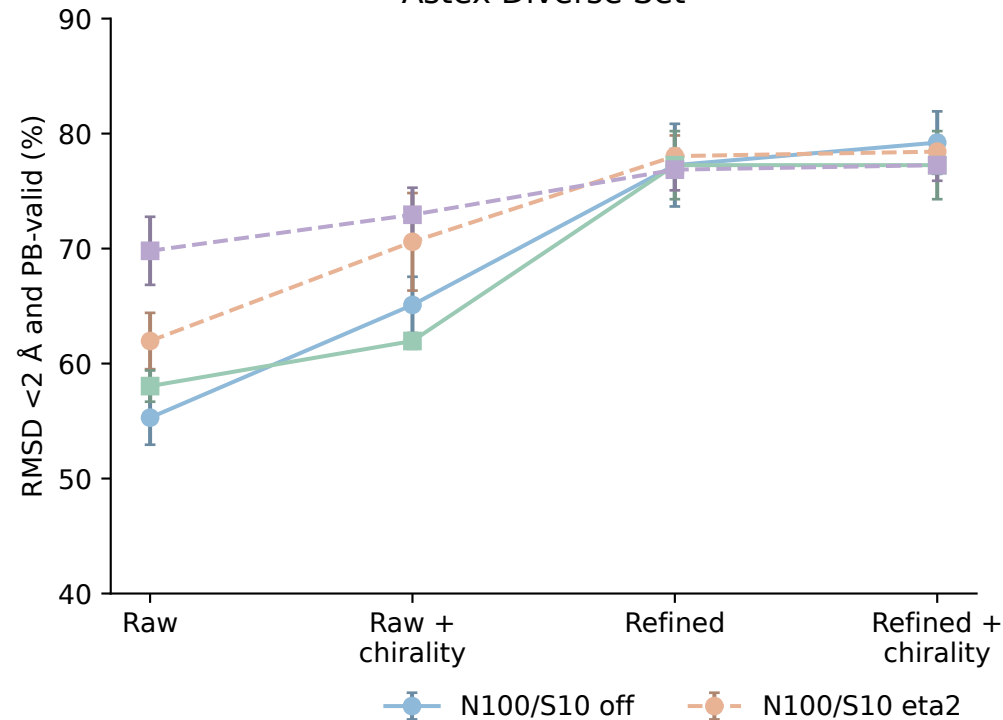


B Generation-order poses

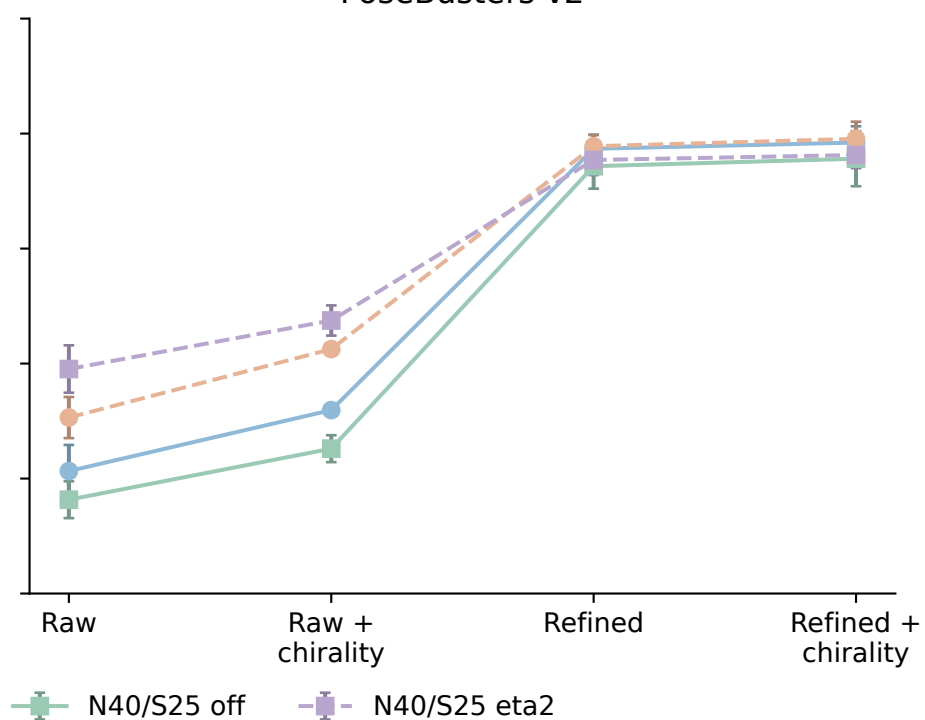


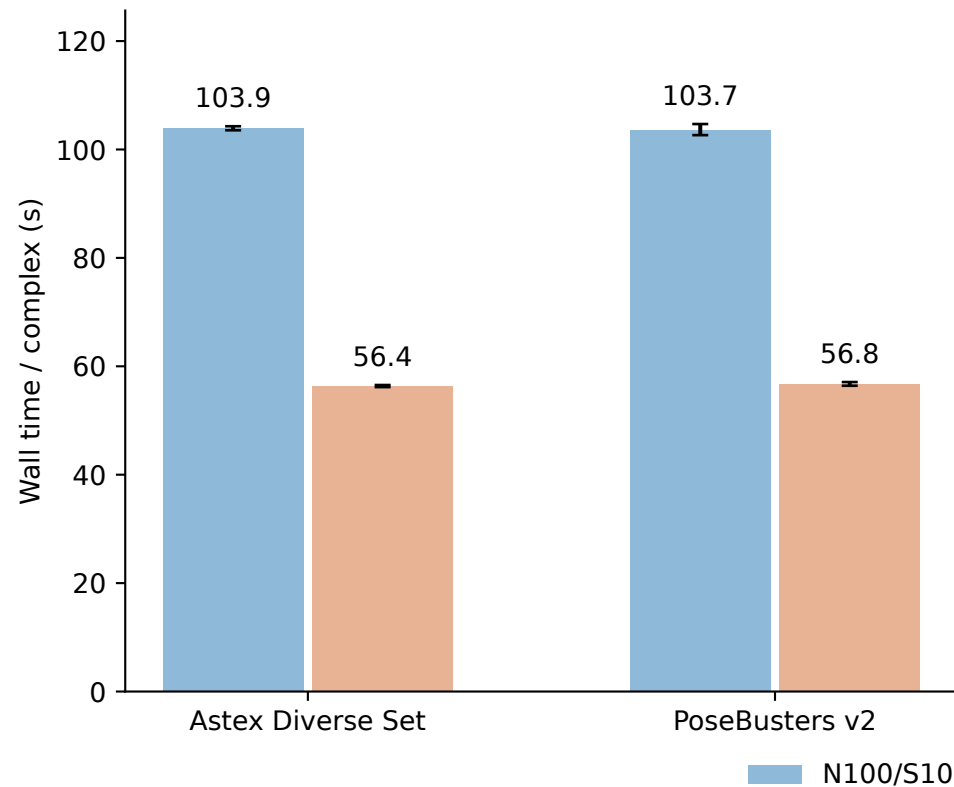
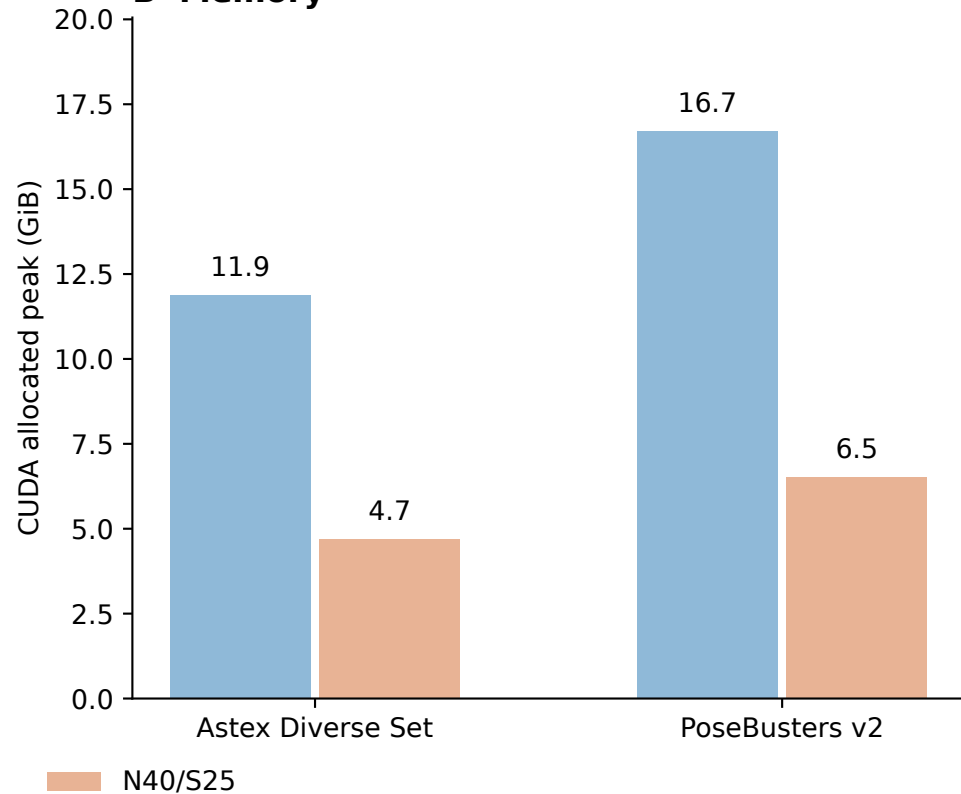


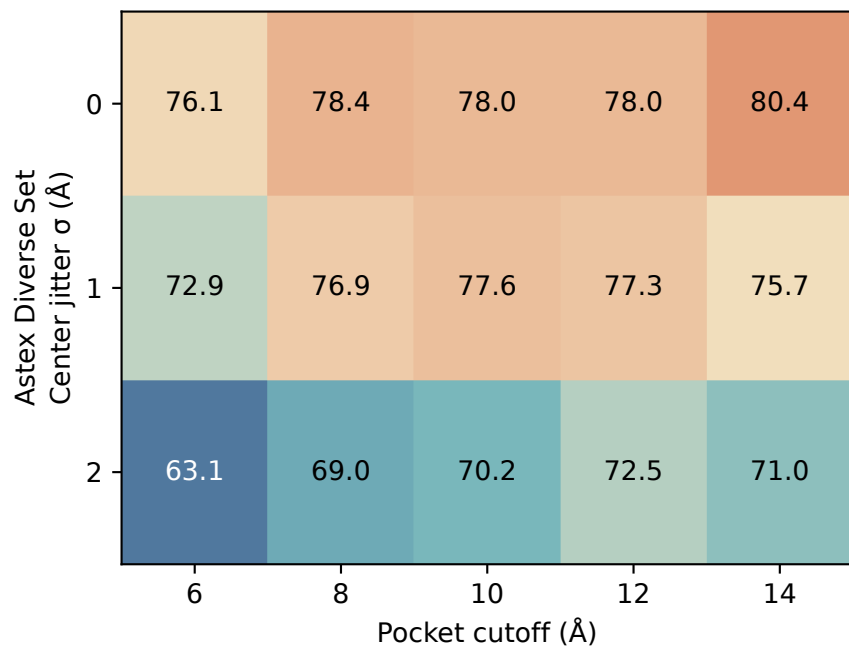
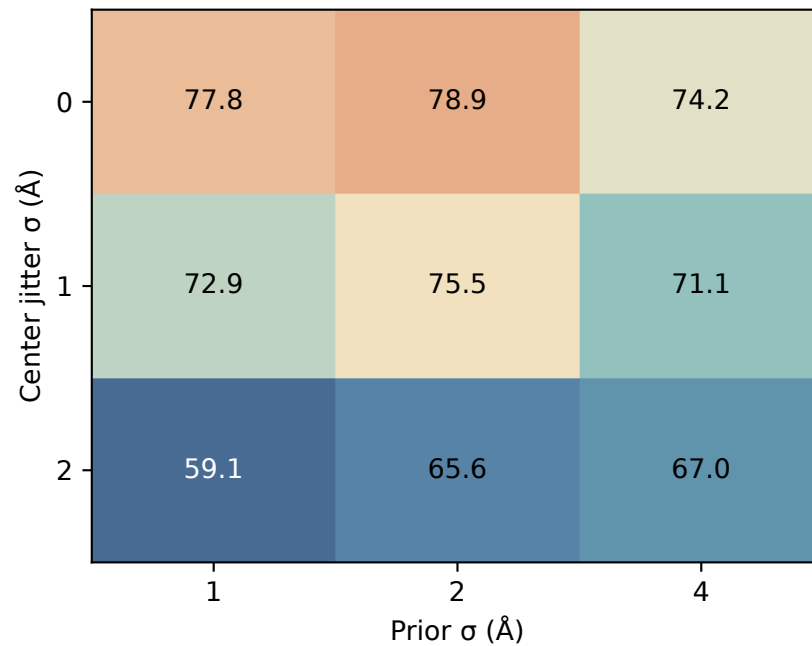
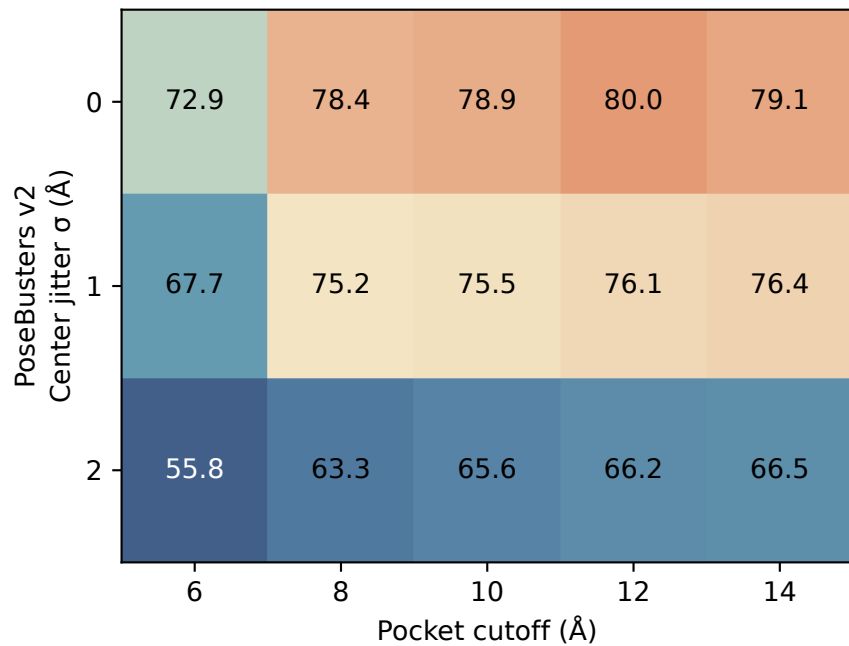
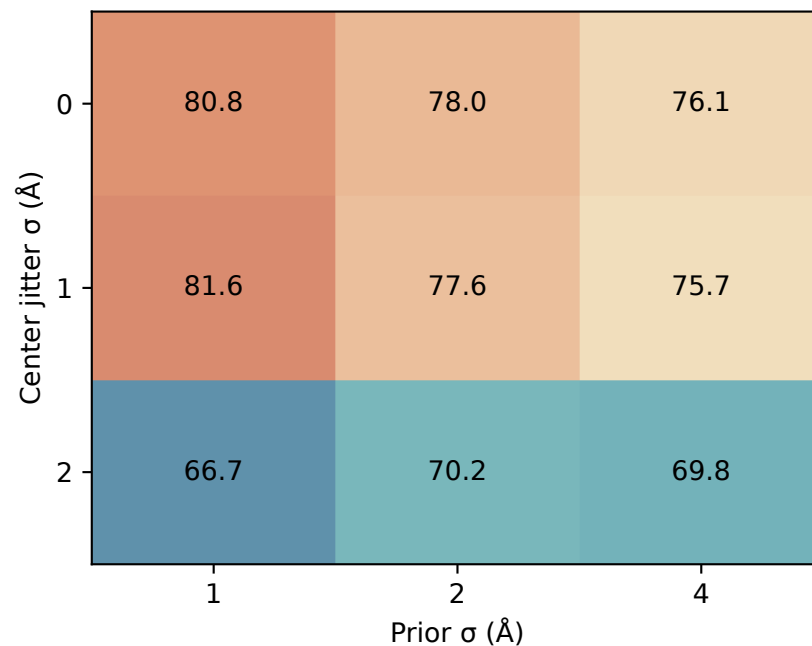
Astex Diverse Set



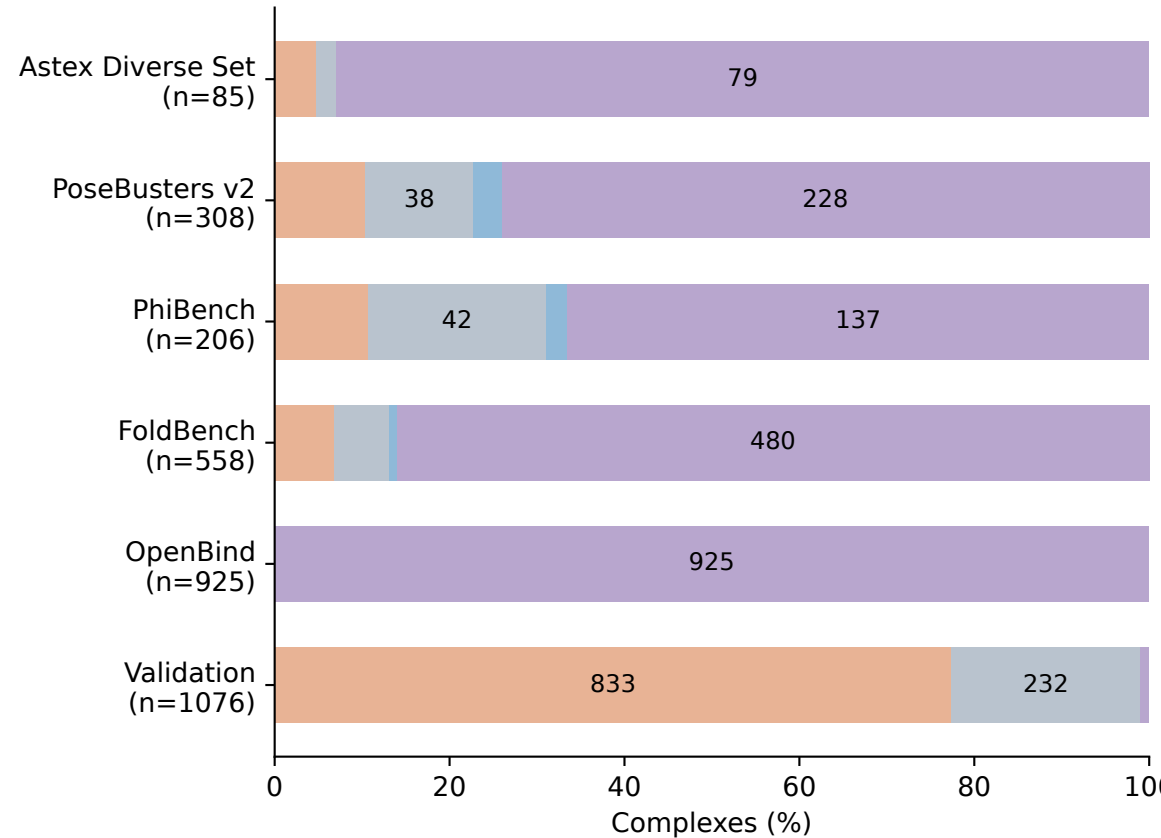
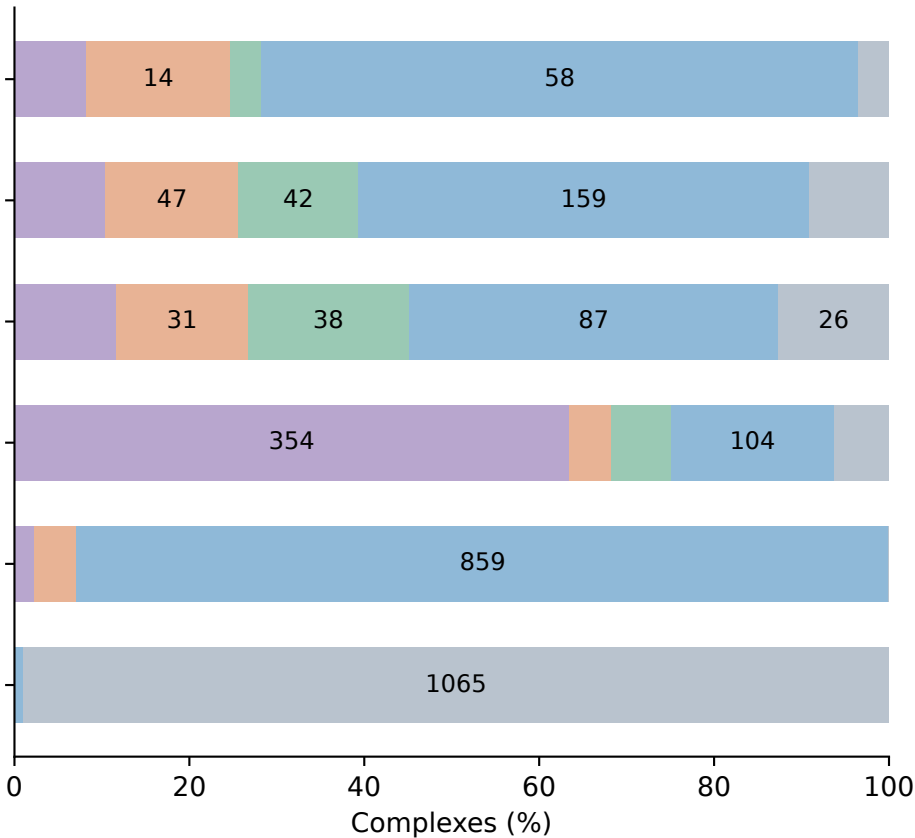
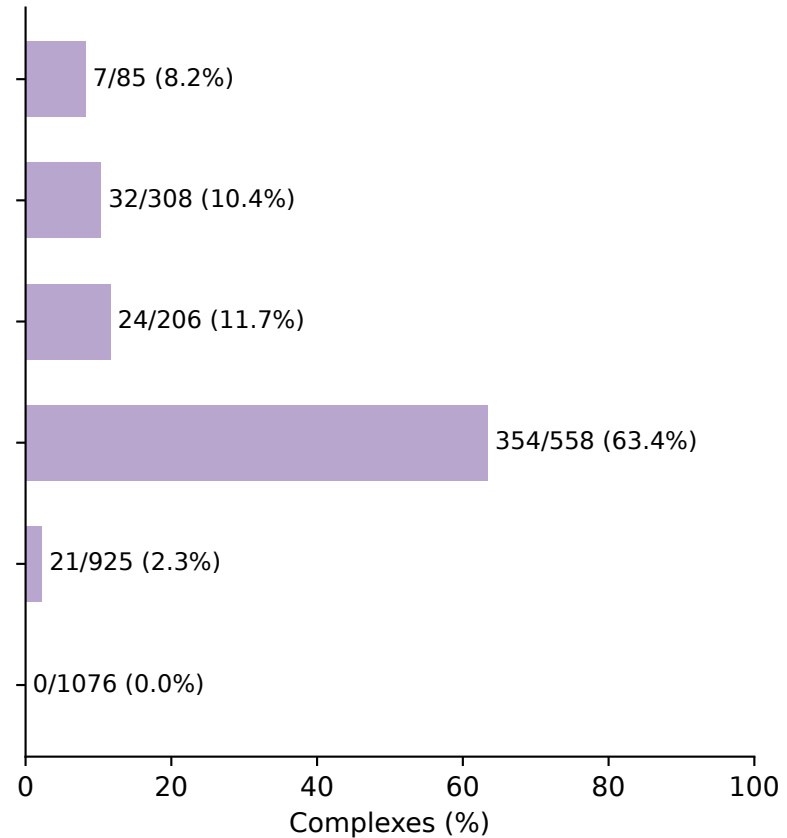
PoseBusters v2



A Runtime**B Memory**

A Pocket size**B Prior scale**

RMSD < 2 Å & PB-valid (%)

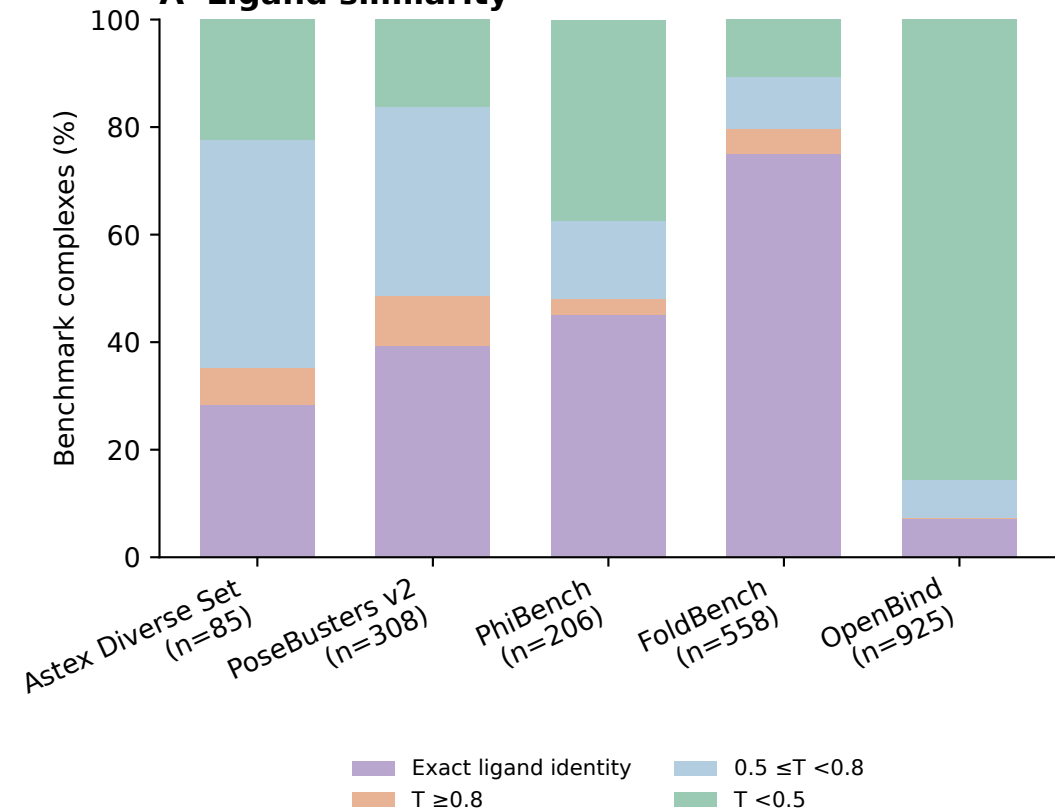
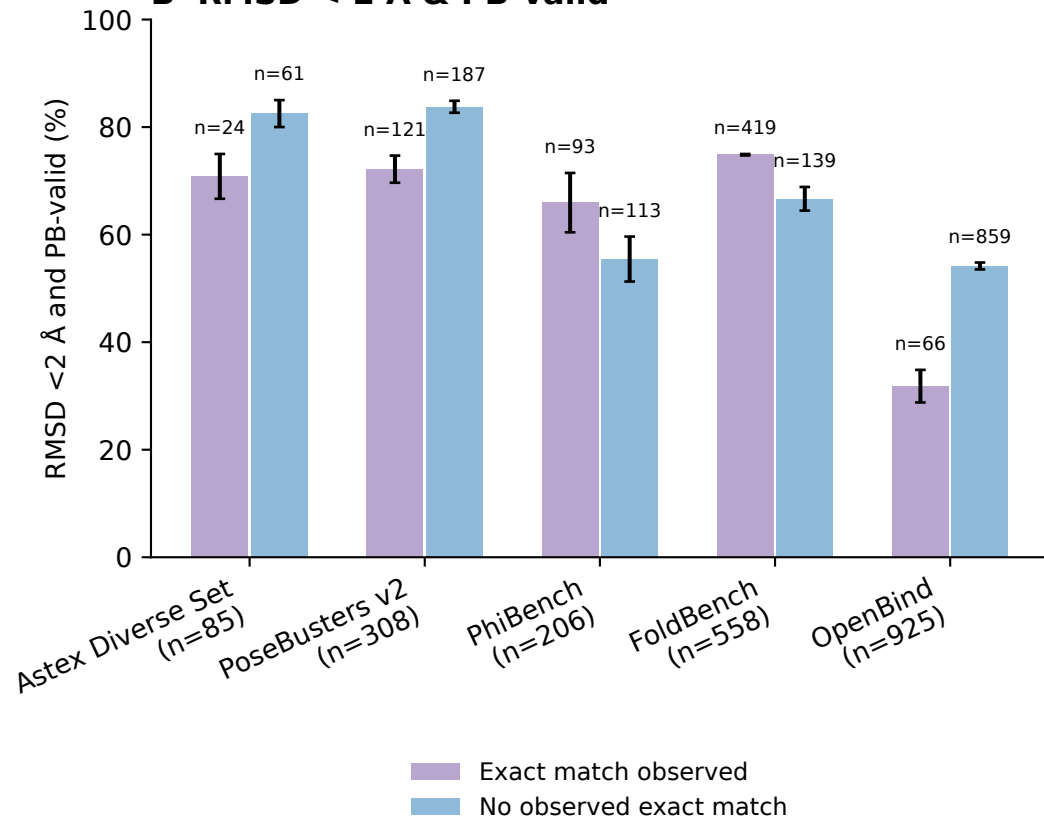
A Sequence identity**B Ligand-protein overlap****C Same training sample**

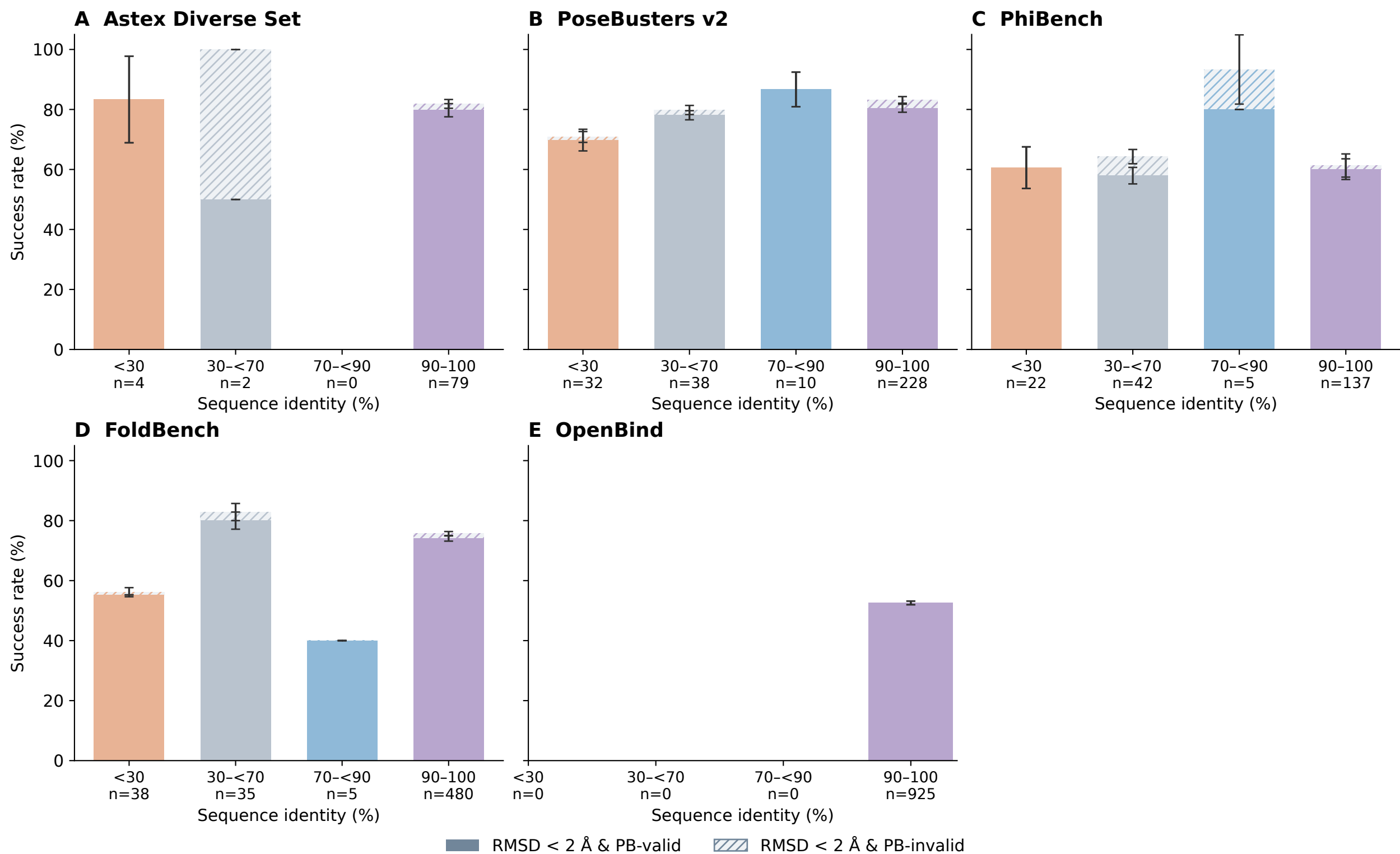
Sequence identity (%)

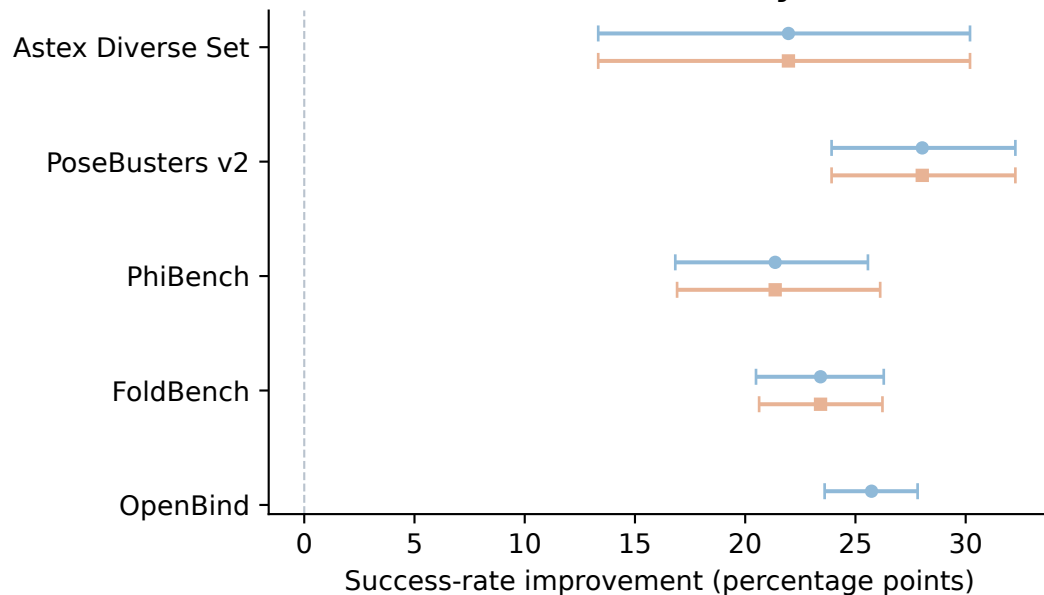
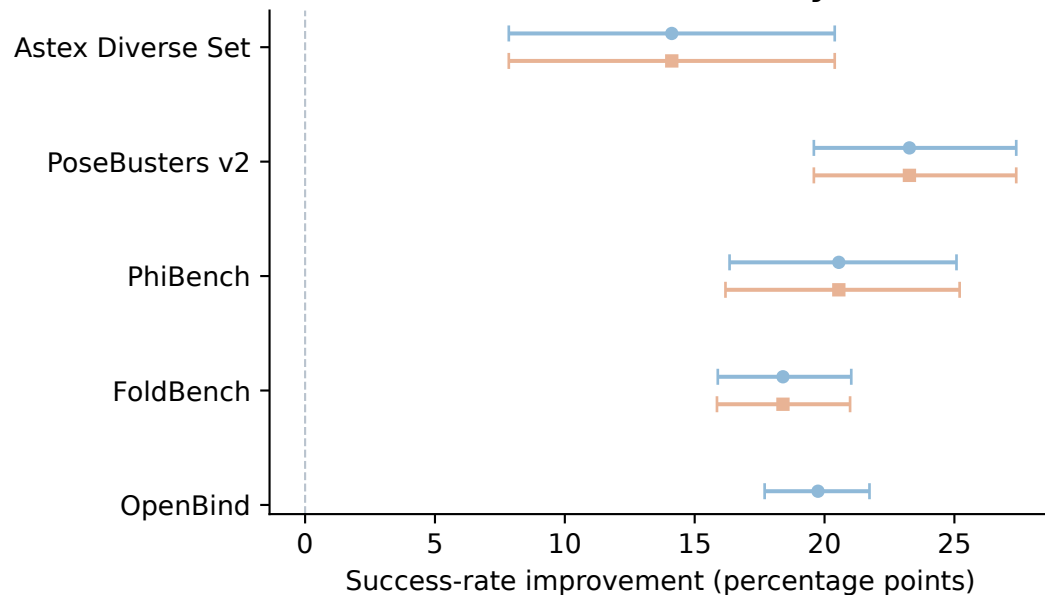
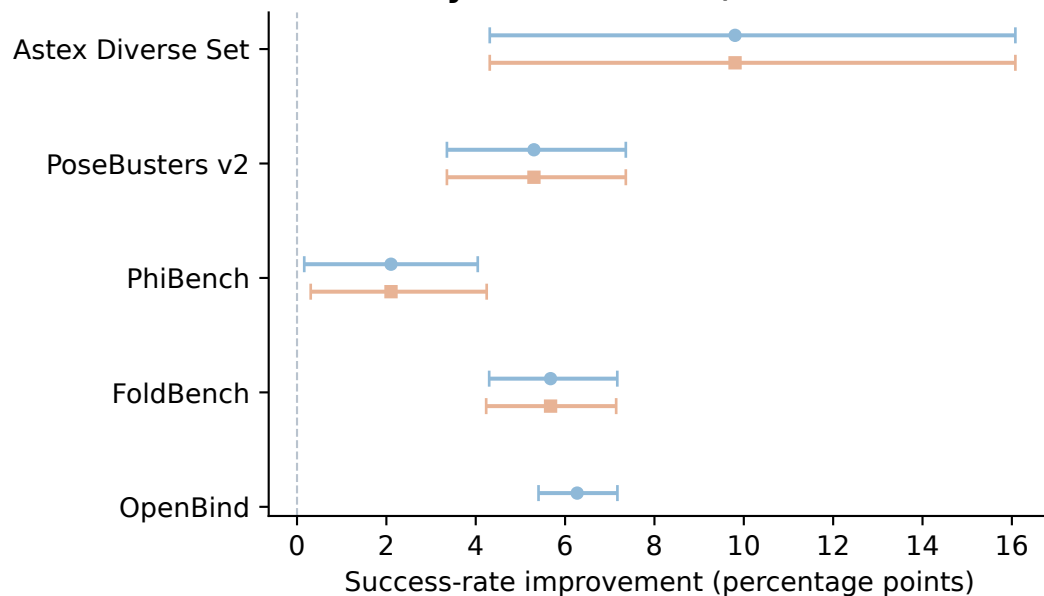
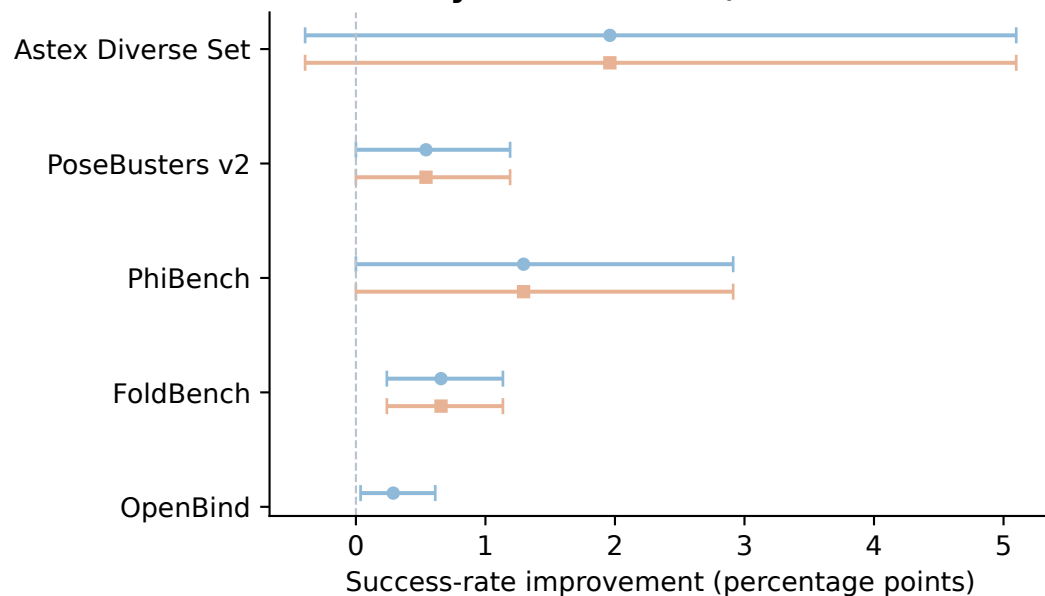
- <30%
- 30-70%
- 70-90%
- 90-100%

Ligand-protein overlap

- Same-sample overlap
- Separate matches
- Ligand only
- Sequence only
- Neither observed

A Ligand similarity**B RMSD < 2 Å & PB-valid**

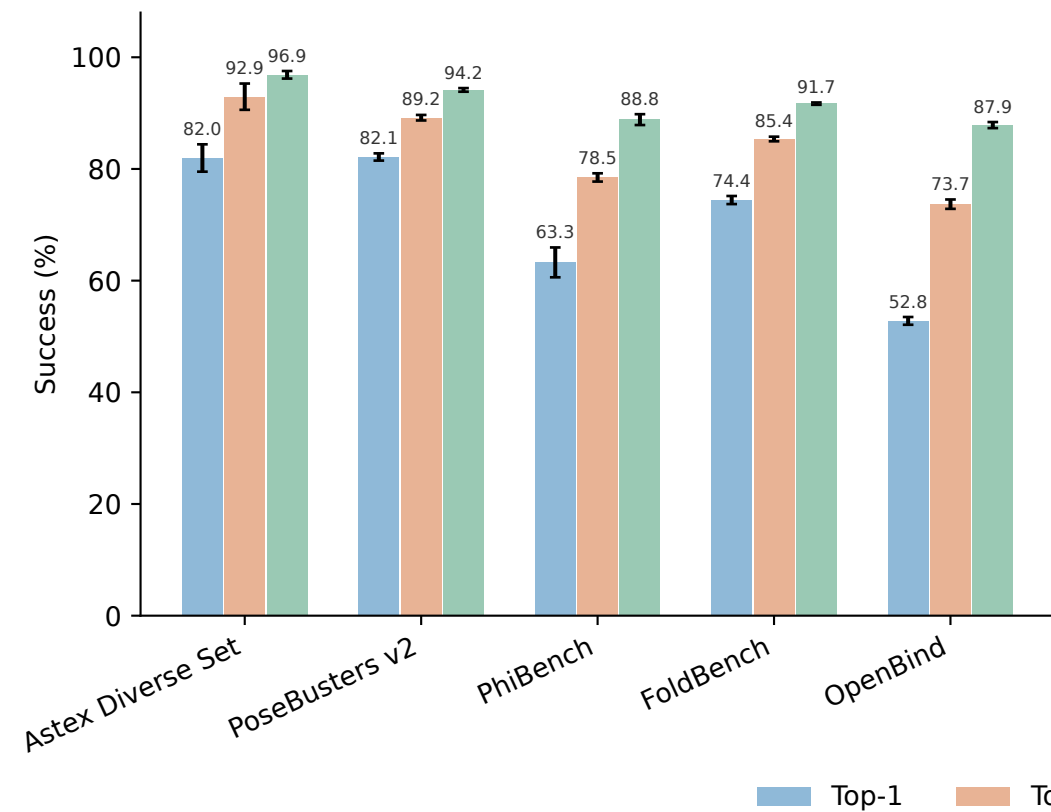
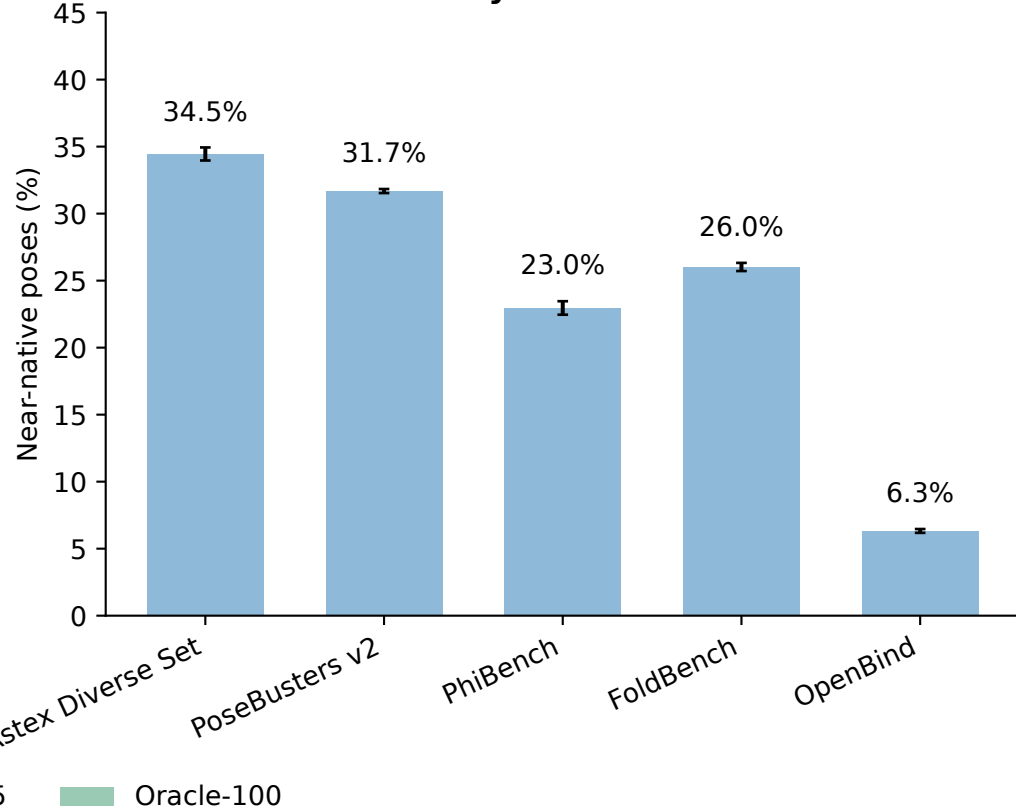


A Refined – raw, no chirality filter**B Refined – raw, with chirality filter****C Chirality filter off → on, raw****D Chirality filter off → on, refined**

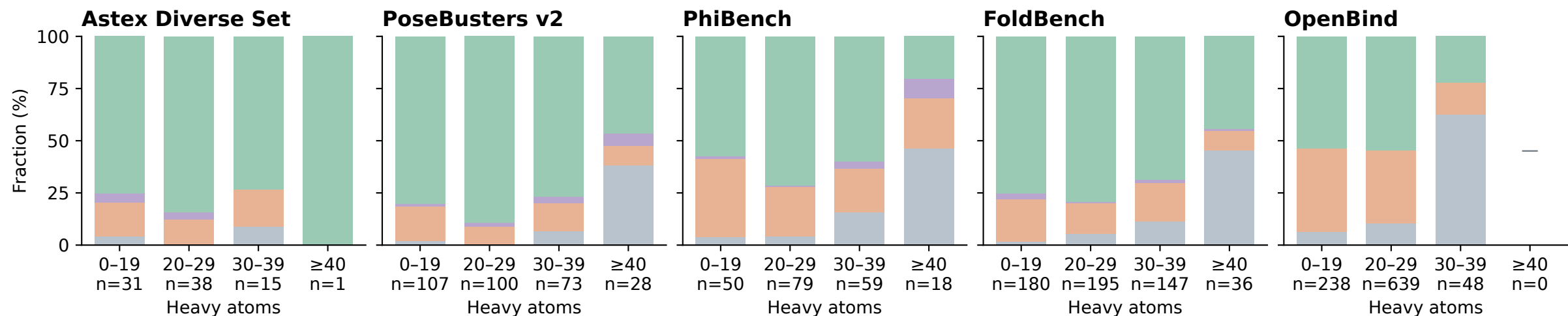
Complex bootstrap



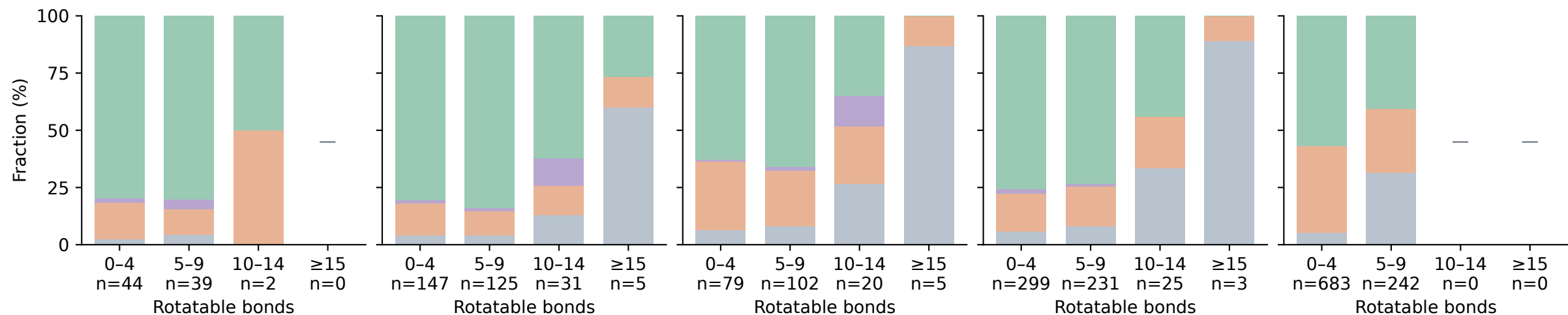
PDB-cluster bootstrap

A Selection**B Near-native density**

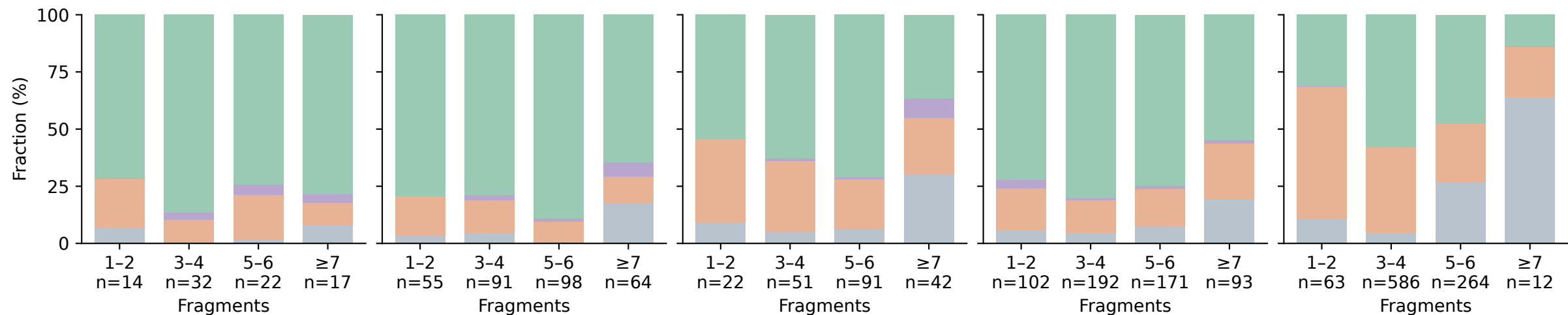
A Ligand size dependence



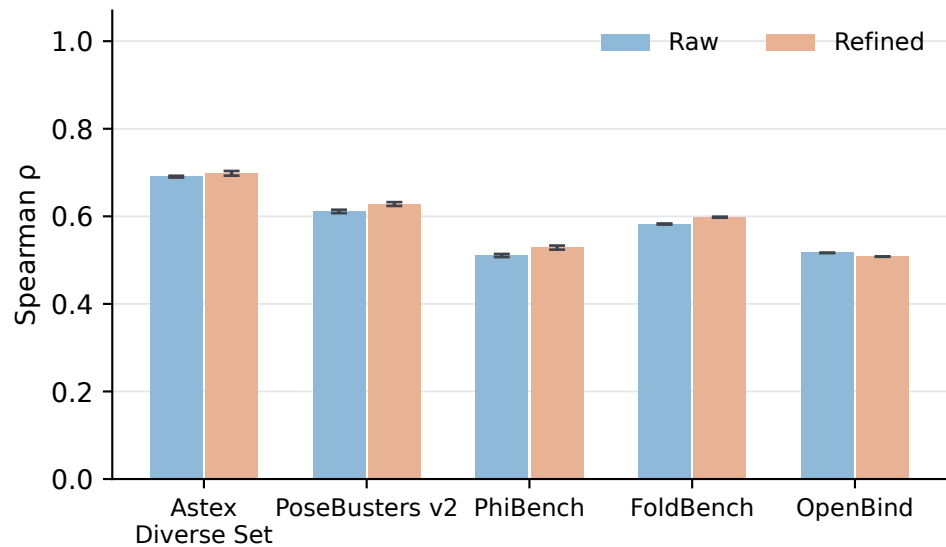
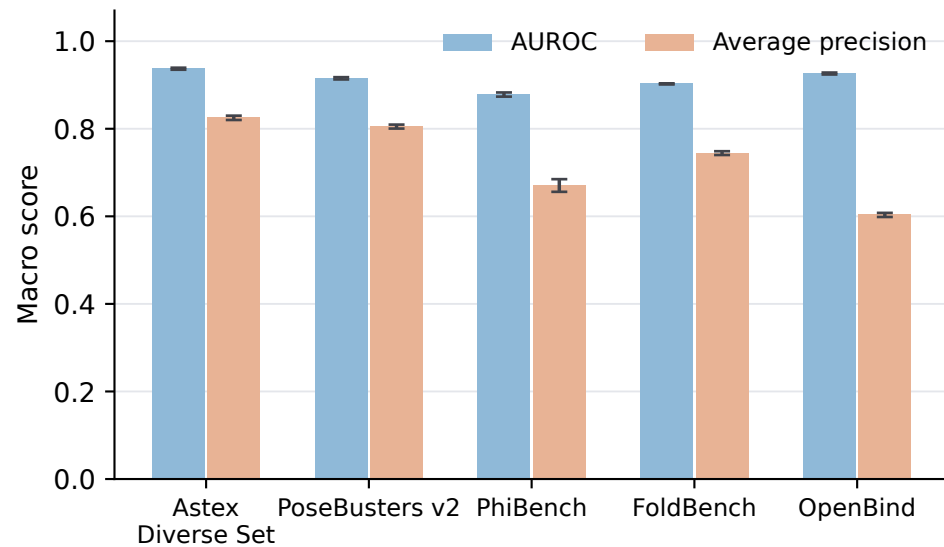
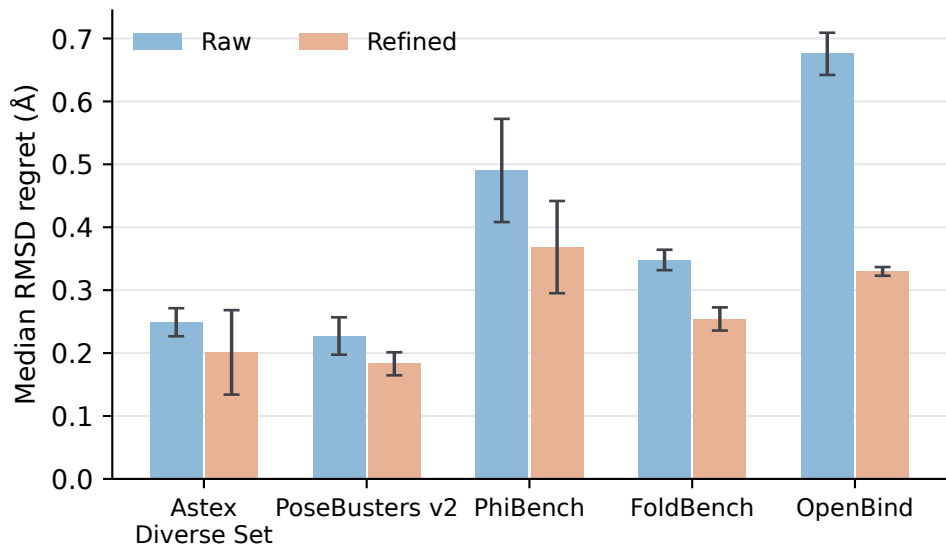
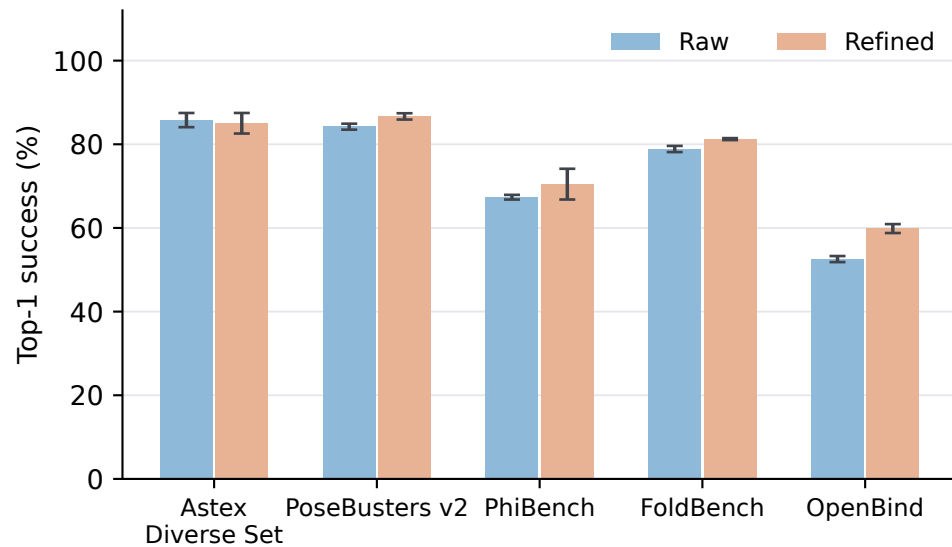
B Rotatable bond dependence

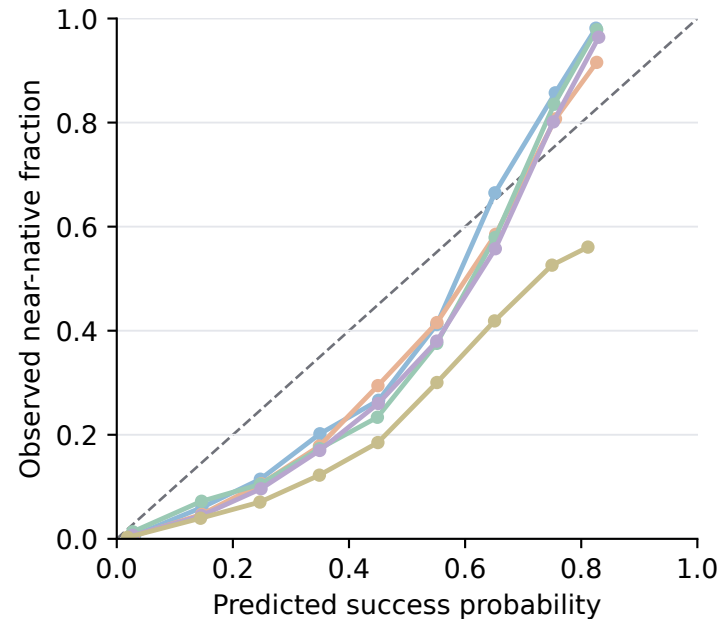
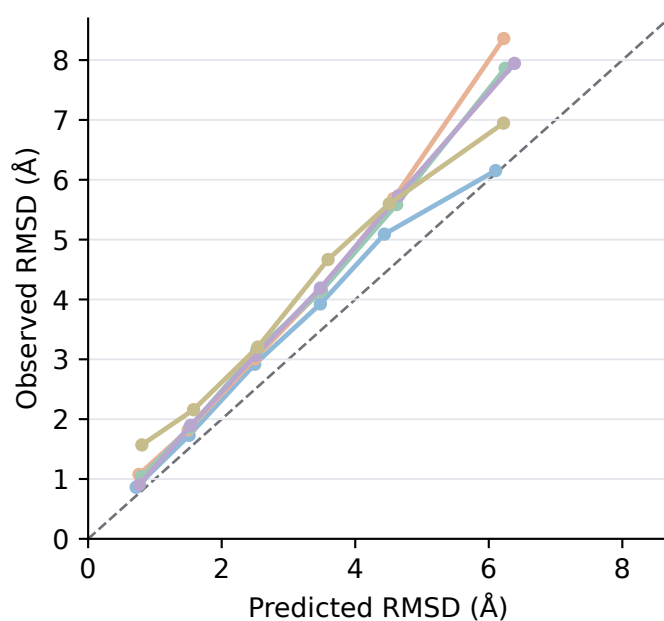
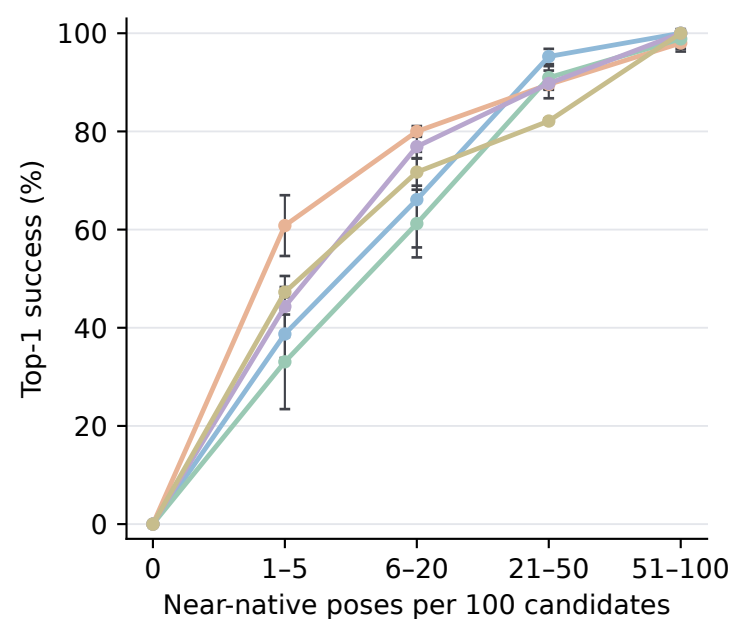


C Fragment count dependence

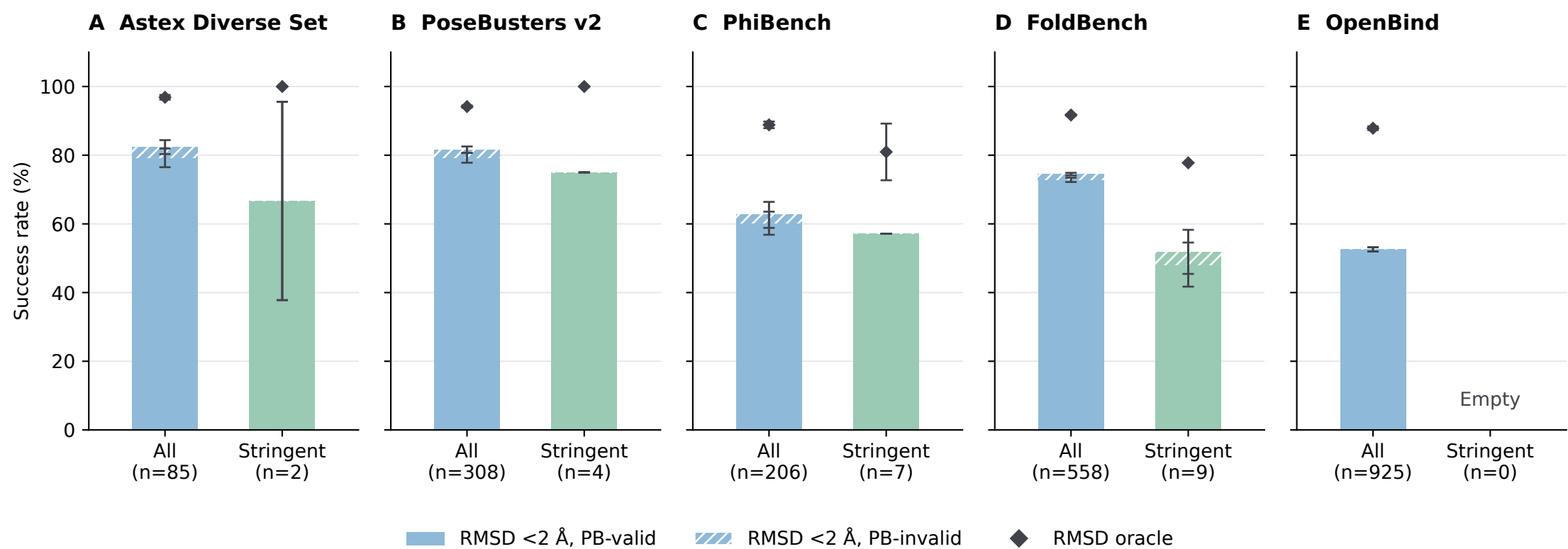


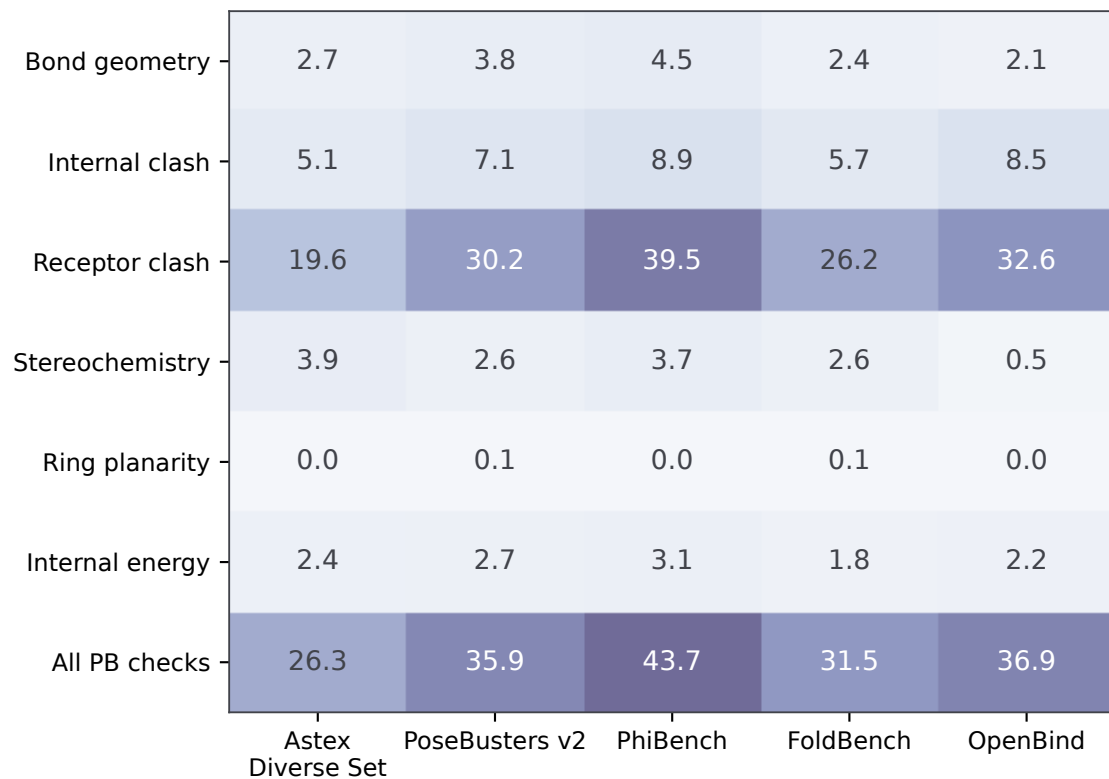
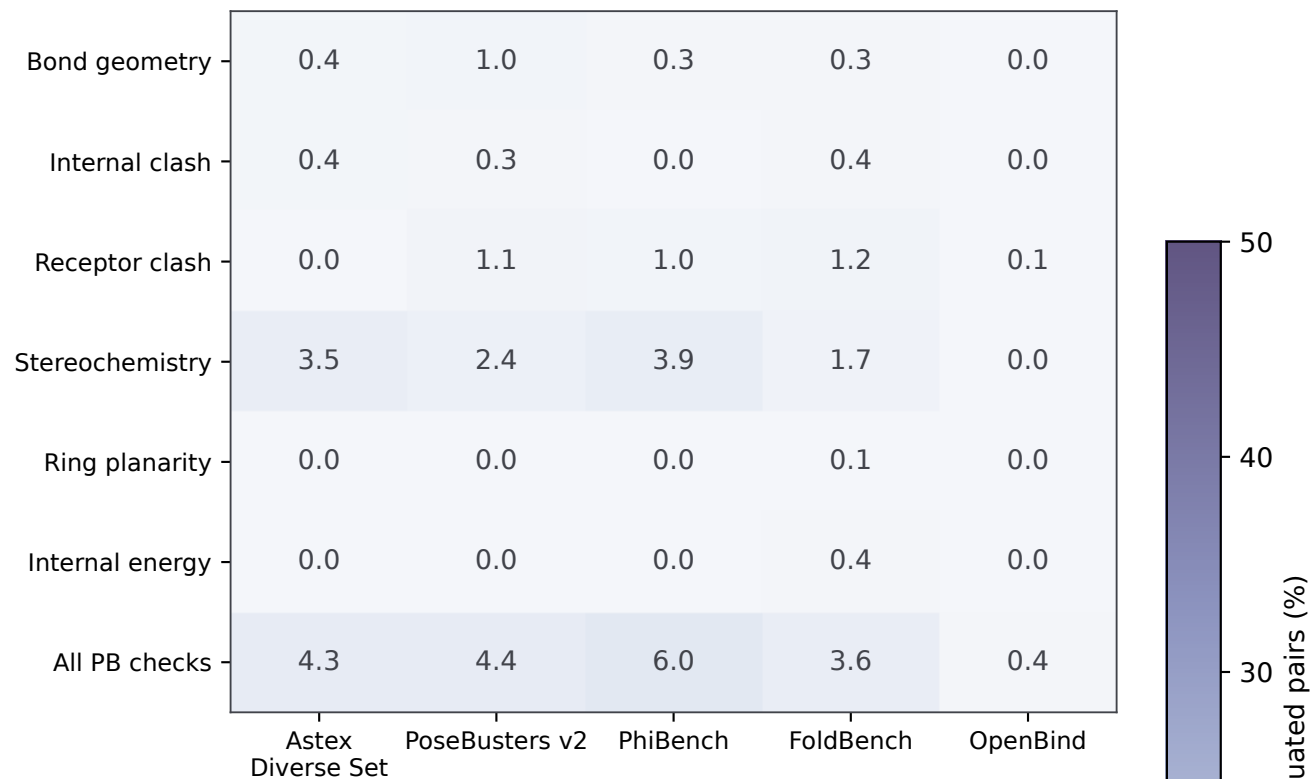
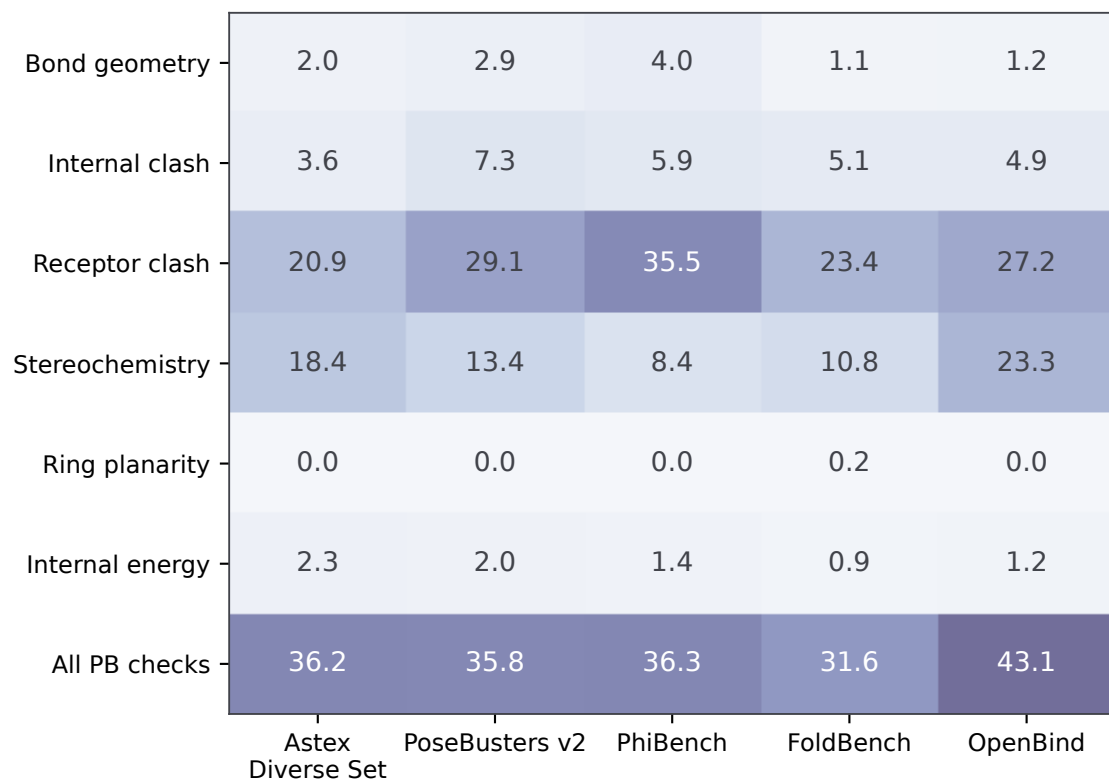
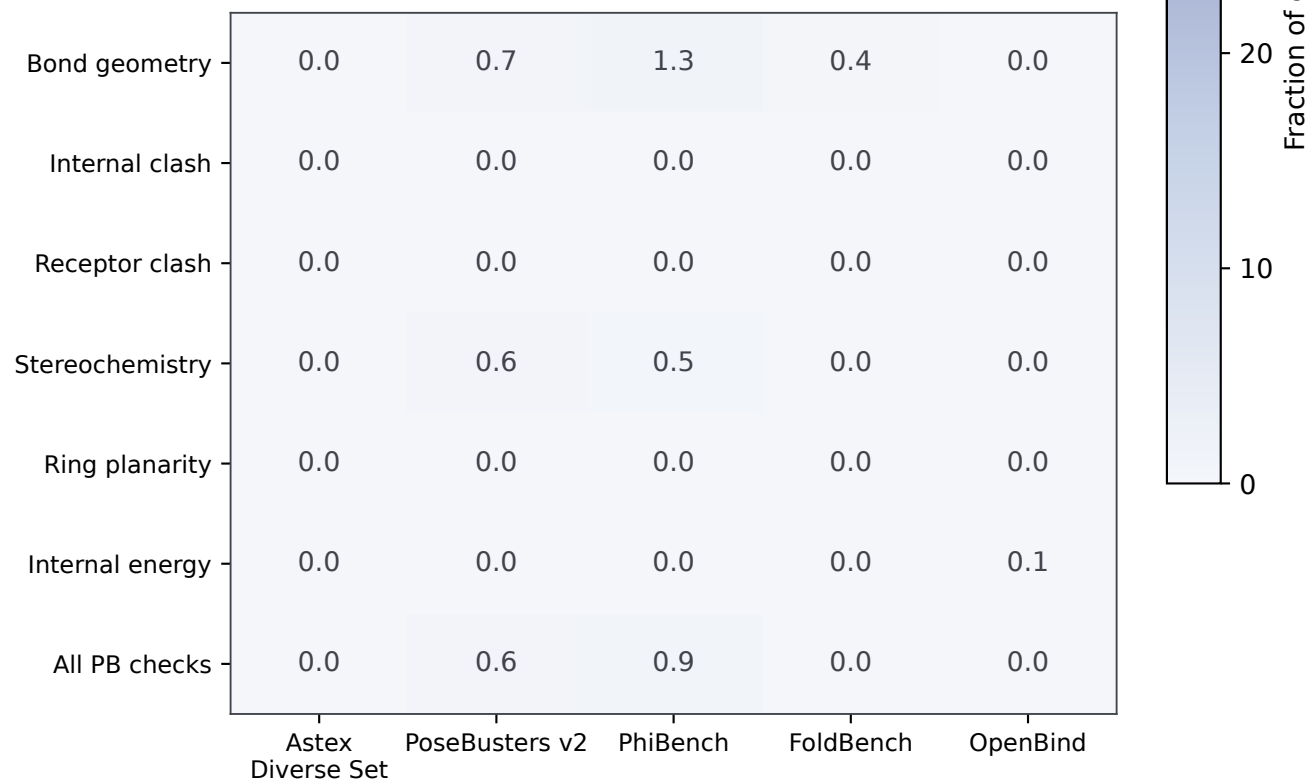
■ Near-native pose generation failure ■ Selection failure ■ PB failure ■ Success

A Within-bank ranking**B Near-native discrimination****C Selection regret****D Selection given a successful candidate**

A Success-probability reliability**B RMSD prediction reliability****C Near-native candidate density**

—●— Astex Diverse Set —●— PoseBusters v2 —●— PhiBench —●— FoldBench —●— OpenBind



A Raw selected-pose failure**B Refined selected-pose failure****C Matched candidate: fail → pass****D Matched candidate: pass → fail**

Fraction of evaluated pairs (%)

50

40

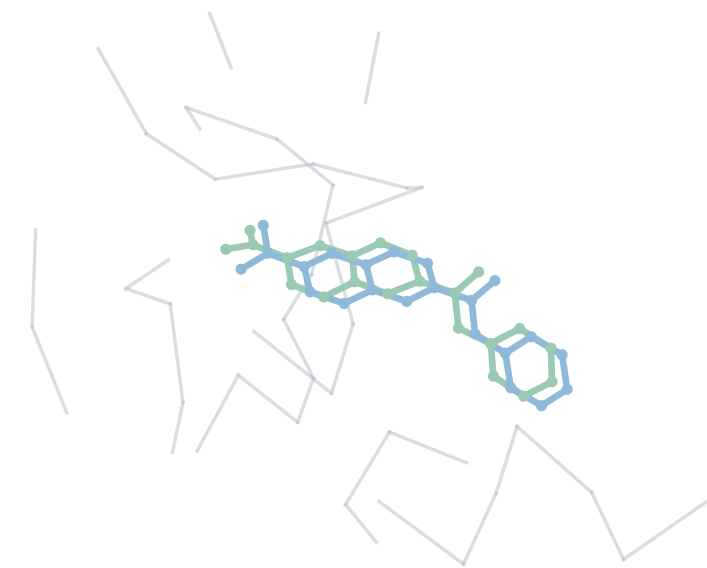
30

20

10

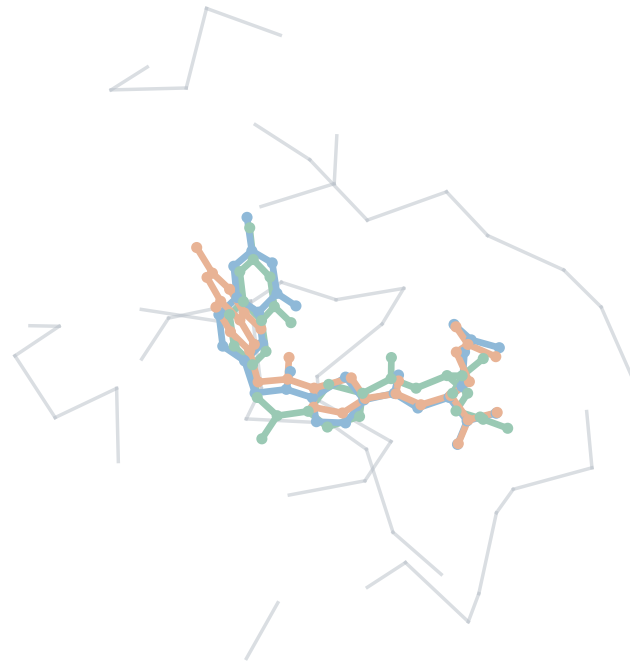
0

A Successful pose
1OWE



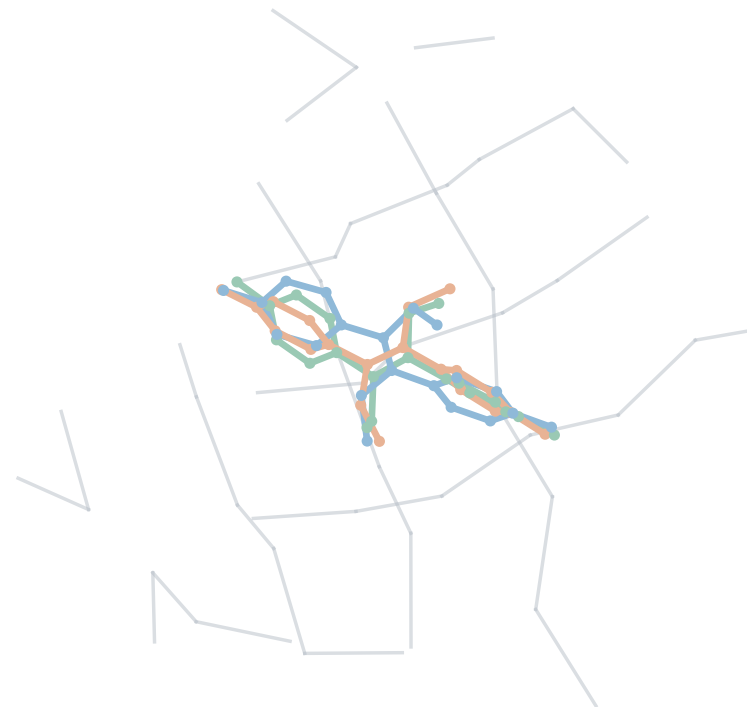
Selected: 0.80 Å

B Refinement rescue
1HVY



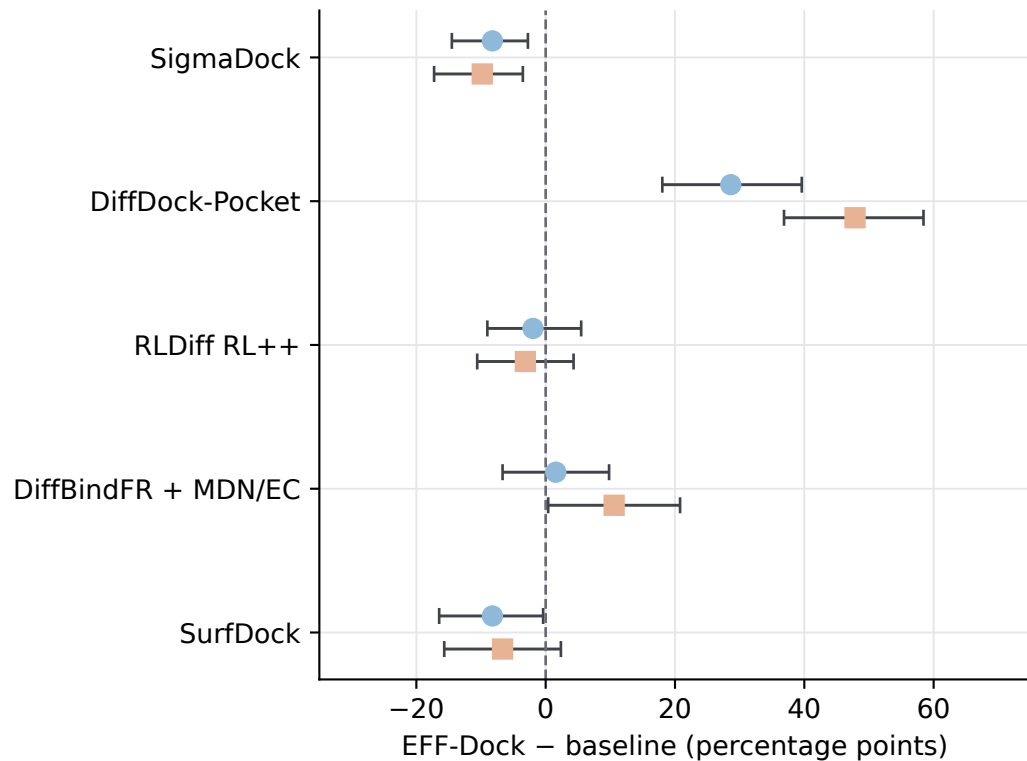
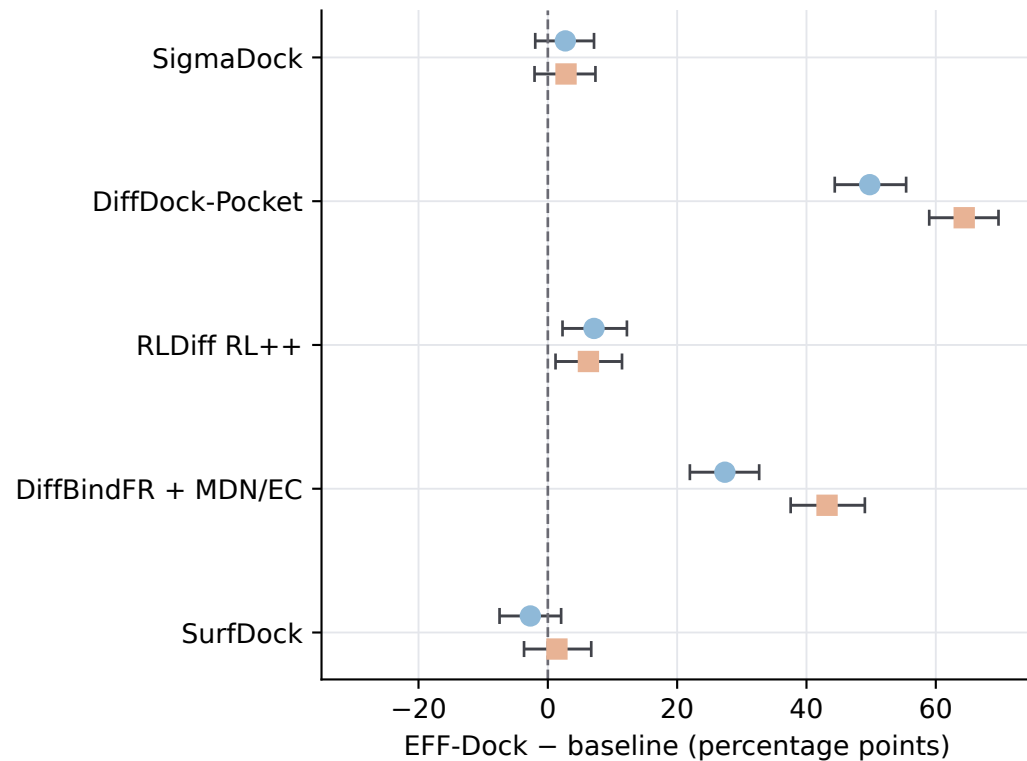
Selected: 1.98 Å · Raw, same pose: 2.32 Å

C Selection failure
1TZ8



Selected: 2.27 Å · Oracle: 0.65 Å

— Crystal ligand — Selected pose — Raw pose / oracle — Protein Cα trace

A Astex Diverse Set**B PoseBusters v2**

● RMSD success ■ PB-valid success